

Lecture Notes: Fletcher–Reeves Method (Conjugate Gradient)

1 Motivation

The Conjugate Gradient method is an efficient alternative to steepest descent for minimizing quadratic functions of the form

$$f(x) = \frac{1}{2}x^\top Ax + b^\top x + c,$$

where A is symmetric positive definite. Unlike steepest descent, which may zig-zag, the Conjugate Gradient method constructs mutually A -conjugate directions, ensuring convergence in at most n iterations for quadratic functions.

2 Line Minimization

Consider a quadratic objective function of the form

$$f(x) = \frac{1}{2}x^\top Ax + b^\top x + c,$$

where A is symmetric (and, no contexto do método, definida positiva).

Given a current point x_i and a search direction S_i , we define the next iterate as

$$x_{i+1} = x_i + \lambda_i S_i,$$

where the scalar λ_i is chosen to minimize f along the line

$$\phi(\lambda) = f(x_i + \lambda S_i).$$

Step 1: Gradient of the quadratic function

For the quadratic function above, the gradient is

$$\nabla f(x) = Ax + b.$$

Em particular, no ponto x_i temos

$$\nabla f_i := \nabla f(x_i) = Ax_i + b.$$

Step 2: Expressing the gradient along the line

Ao longo da reta $x(\lambda) = x_i + \lambda S_i$, o gradiente em função de λ é

$$\nabla f(x_i + \lambda S_i) = A(x_i + \lambda S_i) + b.$$

Usando a distributividade,

$$\nabla f(x_i + \lambda S_i) = Ax_i + \lambda AS_i + b.$$

Reagrupando os termos conhecidos em x_i :

$$\nabla f(x_i + \lambda S_i) = (Ax_i + b) + \lambda AS_i = \nabla f_i + \lambda AS_i.$$

Step 3: Derivative of the line function $\phi(\lambda)$

A função de uma variável real que estamos minimizando é

$$\phi(\lambda) = f(x_i + \lambda S_i).$$

Pela regra da cadeia,

$$\frac{d\phi}{d\lambda} = \nabla f(x_i + \lambda S_i)^\top \frac{d}{d\lambda}(x_i + \lambda S_i).$$

Como

$$\frac{d}{d\lambda}(x_i + \lambda S_i) = S_i,$$

temos

$$\frac{d\phi}{d\lambda} = \nabla f(x_i + \lambda S_i)^\top S_i.$$

Substituindo a expressão da Subsecção 2:

$$\frac{d\phi}{d\lambda} = (\nabla f_i + \lambda AS_i)^\top S_i.$$

Step 4: Simplifying the derivative

Usando $(u + v)^\top S_i = u^\top S_i + v^\top S_i$:

$$\frac{d\phi}{d\lambda} = \nabla f_i^\top S_i + (\lambda A S_i)^\top S_i.$$

Note que $(\lambda A S_i)^\top S_i = \lambda (A S_i)^\top S_i$. Como A é simétrica, $(A S_i)^\top S_i = S_i^\top A S_i$. Logo:

$$\frac{d\phi}{d\lambda} = \nabla f_i^\top S_i + \lambda S_i^\top A S_i.$$

Podemos também escrever

$$\frac{d\phi}{d\lambda} = S_i^\top \nabla f_i + \lambda S_i^\top A S_i.$$

Step 5: Optimal step size

O valor ótimo λ_i^* é obtido impondo a condição de primeira ordem

$$\frac{d\phi}{d\lambda} = 0.$$

Assim,

$$S_i^\top \nabla f_i + \lambda_i^* S_i^\top A S_i = 0.$$

Isolando λ_i^* :

$$\begin{aligned} \lambda_i^* S_i^\top A S_i &= -S_i^\top \nabla f_i, \\ \lambda_i^* &= -\frac{S_i^\top \nabla f_i}{S_i^\top A S_i}. \end{aligned}$$

Em particular, na primeira iteração ($i = 1$) temos

$$\lambda_1^* = -\frac{S_1^\top \nabla f_1}{S_1^\top A S_1}.$$

3 Conjugate Directions

We now show in detail how to construct the second search direction S_2 and how to obtain the scalar β_2 that makes S_1 and S_2 A -conjugate.

Step 1: First search direction

The first search direction is chosen as the negative gradient:

$$S_1 = -\nabla f_1,$$

where

$$\nabla f_1 := \nabla f(x_1).$$

Step 2: Second search direction as a combination

We assume that the second direction is a linear combination of the new steepest descent direction $-\nabla f_2$ and the previous direction S_1 :

$$S_2 = -\nabla f_2 + \beta_2 S_1,$$

where

$$\nabla f_2 := \nabla f(x_2), \quad x_2 = x_1 + \lambda_1^* S_1,$$

and β_2 is an unknown scalar to be determined.

Step 3: Conjugacy condition

We want S_1 and S_2 to be A -conjugate, i.e.,

$$S_1^\top A S_2 = 0.$$

Substitute the expression of S_2 :

$$S_1^\top A (-\nabla f_2 + \beta_2 S_1) = 0.$$

Distribute A and separate the terms:

$$-S_1^\top A \nabla f_2 + \beta_2 S_1^\top A S_1 = 0.$$

Now solve for β_2 :

$$\begin{aligned} \beta_2 S_1^\top A S_1 &= S_1^\top A \nabla f_2, \\ \beta_2 &= \frac{S_1^\top A \nabla f_2}{S_1^\top A S_1}. \end{aligned}$$

Neste ponto, a expressão ainda contém a matriz A . O próximo passo é reescrevê-la usando apenas gradientes.

Step 4: Relation between consecutive gradients

Recall that, for the quadratic function

$$f(x) = \frac{1}{2}x^\top Ax + b^\top x + c,$$

the gradient is

$$\nabla f(x) = Ax + b.$$

At x_1 and x_2 :

$$\nabla f_1 = Ax_1 + b, \quad \nabla f_2 = Ax_2 + b.$$

Since

$$x_2 = x_1 + \lambda_1^* S_1,$$

we have

$$\nabla f_2 = A(x_1 + \lambda_1^* S_1) + b = Ax_1 + \lambda_1^* AS_1 + b = (Ax_1 + b) + \lambda_1^* AS_1 = \nabla f_1 + \lambda_1^* AS_1.$$

Rearranging, we obtain:

$$AS_1 = \frac{\nabla f_2 - \nabla f_1}{\lambda_1^*}.$$

Step 5: Denominator $S_1^\top AS_1$ in terms of gradients

Use the identity above:

$$S_1^\top AS_1 = S_1^\top \left(\frac{\nabla f_2 - \nabla f_1}{\lambda_1^*} \right) = \frac{S_1^\top \nabla f_2 - S_1^\top \nabla f_1}{\lambda_1^*}.$$

Next, we will use the fact that the line search is exact.

Step 6: Orthogonality from exact line search

Consider the line function

$$\phi(\lambda) = f(x_1 + \lambda S_1).$$

We know that the derivative at the optimal step λ_1^* is zero:

$$\left. \frac{d\phi}{d\lambda} \right|_{\lambda=\lambda_1^*} = 0.$$

As seen before,

$$\frac{d\phi}{d\lambda} = \nabla f(x_1 + \lambda S_1)^\top S_1,$$

thus, at $\lambda = \lambda_1^*$,

$$\nabla f_2^\top S_1 = 0 \implies S_1^\top \nabla f_2 = 0.$$

Substitute this into the denominator:

$$S_1^\top AS_1 = \frac{0 - S_1^\top \nabla f_1}{\lambda_1^*} = -\frac{S_1^\top \nabla f_1}{\lambda_1^*}.$$

Using $S_1 = -\nabla f_1$:

$$S_1^\top \nabla f_1 = -\nabla f_1^\top \nabla f_1,$$

so

$$S_1^\top AS_1 = -\frac{-\nabla f_1^\top \nabla f_1}{\lambda_1^*} = \frac{\nabla f_1^\top \nabla f_1}{\lambda_1^*}.$$

Step 7: Numerator $S_1^\top A \nabla f_2$ in terms of gradients

Using $AS_1 = (\nabla f_2 - \nabla f_1)/\lambda_1^*$:

$$S_1^\top A \nabla f_2 = (AS_1)^\top \nabla f_2 = \left(\frac{\nabla f_2 - \nabla f_1}{\lambda_1^*} \right)^\top \nabla f_2.$$

Compute the product:

$$S_1^\top A \nabla f_2 = \frac{\nabla f_2^\top \nabla f_2 - \nabla f_1^\top \nabla f_2}{\lambda_1^*}.$$

Agora usamos novamente uma ortogonalidade importante.

Note that

$$S_1 = -\nabla f_1 \implies S_1^\top \nabla f_2 = -\nabla f_1^\top \nabla f_2.$$

But from the exact line search condition we already know that

$$S_1^\top \nabla f_2 = 0 \implies \nabla f_1^\top \nabla f_2 = 0.$$

Thus,

$$S_1^\top A \nabla f_2 = \frac{\nabla f_2^\top \nabla f_2 - 0}{\lambda_1^*} = \frac{\nabla f_2^\top \nabla f_2}{\lambda_1^*}.$$

Step 8: Final expression for β_2

Recall

$$\beta_2 = \frac{S_1^\top A \nabla f_2}{S_1^\top A S_1}.$$

Substitute numerator and denominator:

$$\beta_2 = \frac{\frac{\nabla f_2^\top \nabla f_2}{\lambda_1^*}}{\frac{\nabla f_1^\top \nabla f_1}{\lambda_1^*}} = \frac{\nabla f_2^\top \nabla f_2}{\nabla f_1^\top \nabla f_1}.$$

This is exactly the Fletcher–Reeves formula for β_2 .

4 General Formula for β_i

We now generalize the previous derivation to obtain the expression for β_i for $i \geq 2$.

Step 1: General search direction

Assume that, at iteration i , the search direction is defined by

$$S_i = -\nabla f_i + \beta_i S_{i-1},$$

where $\nabla f_i := \nabla f(x_i)$.

We require S_{i-1} and S_i to be A -conjugate:

$$S_{i-1}^\top A S_i = 0.$$

Substitute S_i :

$$S_{i-1}^\top A (-\nabla f_i + \beta_i S_{i-1}) = 0.$$

Expand:

$$-S_{i-1}^\top A \nabla f_i + \beta_i S_{i-1}^\top A S_{i-1} = 0.$$

Solve for β_i :

$$\begin{aligned} \beta_i S_{i-1}^\top A S_{i-1} &= S_{i-1}^\top A \nabla f_i, \\ \beta_i &= \frac{S_{i-1}^\top A \nabla f_i}{S_{i-1}^\top A S_{i-1}}. \end{aligned}$$

Again, we want to eliminate A using gradient relations.

Step 2: Relation between ∇f_i and ∇f_{i-1}

As before, for the quadratic function

$$\nabla f(x) = Ax + b.$$

At iterations $i-1$ and i :

$$\nabla f_{i-1} = Ax_{i-1} + b, \quad \nabla f_i = Ax_i + b.$$

We also have

$$x_i = x_{i-1} + \lambda_{i-1}^* S_{i-1}.$$

Therefore,

$$\nabla f_i = A(x_{i-1} + \lambda_{i-1}^* S_{i-1}) + b = (Ax_{i-1} + b) + \lambda_{i-1}^* A S_{i-1} = \nabla f_{i-1} + \lambda_{i-1}^* A S_{i-1}.$$

Rearranging:

$$A S_{i-1} = \frac{\nabla f_i - \nabla f_{i-1}}{\lambda_{i-1}^*}.$$

Step 3: Denominator $S_{i-1}^\top AS_{i-1}$

Compute:

$$S_{i-1}^\top AS_{i-1} = S_{i-1}^\top \left(\frac{\nabla f_i - \nabla f_{i-1}}{\lambda_{i-1}^*} \right) = \frac{S_{i-1}^\top \nabla f_i - S_{i-1}^\top \nabla f_{i-1}}{\lambda_{i-1}^*}.$$

As in the case $i = 1$, the exact line search at step $i - 1$ implies

$$\left. \frac{d}{d\lambda} f(x_{i-1} + \lambda S_{i-1}) \right|_{\lambda=\lambda_{i-1}^*} = 0,$$

which gives

$$\nabla f_i^\top S_{i-1} = 0 \implies S_{i-1}^\top \nabla f_i = 0.$$

Thus,

$$S_{i-1}^\top AS_{i-1} = \frac{0 - S_{i-1}^\top \nabla f_{i-1}}{\lambda_{i-1}^*} = -\frac{S_{i-1}^\top \nabla f_{i-1}}{\lambda_{i-1}^*}.$$

Using the structure of S_{i-1} , one can show that the gradients are mutually orthogonal and, in particular,

$$\nabla f_{i-1}^\top \nabla f_i = 0,$$

and S_{i-1} has a component $-\nabla f_{i-1}$ dominating the product with ∇f_{i-1} . Em termos práticos, obtém-se

$$S_{i-1}^\top AS_{i-1} = \frac{\nabla f_{i-1}^\top \nabla f_{i-1}}{\lambda_{i-1}^*}.$$

(Para uma prova completa, segue-se o mesmo raciocínio recursivo do caso $i = 1$.)

Step 4: Numerator $S_{i-1}^\top A \nabla f_i$

Using $AS_{i-1} = (\nabla f_i - \nabla f_{i-1})/\lambda_{i-1}^*$:

$$S_{i-1}^\top A \nabla f_i = (AS_{i-1})^\top \nabla f_i = \left(\frac{\nabla f_i - \nabla f_{i-1}}{\lambda_{i-1}^*} \right)^\top \nabla f_i.$$

Compute:

$$S_{i-1}^\top A \nabla f_i = \frac{\nabla f_i^\top \nabla f_i - \nabla f_{i-1}^\top \nabla f_i}{\lambda_{i-1}^*}.$$

Again, by orthogonality of consecutive gradients (from exact line search),

$$\nabla f_{i-1}^\top \nabla f_i = 0,$$

so

$$S_{i-1}^\top A \nabla f_i = \frac{\nabla f_i^\top \nabla f_i}{\lambda_{i-1}^*}.$$

Step 5: Final expression for β_i

Now,

$$\beta_i = \frac{S_{i-1}^\top A \nabla f_i}{S_{i-1}^\top AS_{i-1}} = \frac{\frac{\nabla f_i^\top \nabla f_i}{\lambda_{i-1}^*}}{\frac{\nabla f_{i-1}^\top \nabla f_{i-1}}{\lambda_{i-1}^*}} = \frac{\nabla f_i^\top \nabla f_i}{\nabla f_{i-1}^\top \nabla f_{i-1}}.$$

Thus, for all $i \geq 2$,

$$\beta_i = \frac{\nabla f_i^\top \nabla f_i}{\nabla f_{i-1}^\top \nabla f_{i-1}}.$$

This is the Fletcher–Reeves update formula used in the Conjugate Gradient method.

5 Fletcher–Reeves Algorithm (English Version)

1. Choose initial guess x_1 .
2. Compute ∇f_1 and set $S_1 = -\nabla f_1$.
3. For $i = 1, 2, \dots$:

(a) Compute the step size

$$\lambda_i^* = -\frac{S_i^\top \nabla f_i}{S_i^\top A S_i}.$$

(b) Update the iterate

$$x_{i+1} = x_i + \lambda_i^* S_i.$$

(c) Compute the new gradient ∇f_{i+1} .

(d) If $\|\nabla f_{i+1}\|$ is sufficiently small, stop.

(e) Compute the scalar

$$\beta_{i+1} = \frac{\nabla f_{i+1}^\top \nabla f_{i+1}}{\nabla f_i^\top \nabla f_i}.$$

(f) Update the direction

$$S_{i+1} = -\nabla f_{i+1} + \beta_{i+1} S_i.$$

6 Example 6.9

Minimize:

$$f(x_1, x_2) = x_1 - x_2 + 2x_1^2 + 2x_1x_2 + x_2^2,$$

starting from:

$$X_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Iteration 1

The gradient is:

$$\nabla f = \begin{bmatrix} 1 + 4x_1 + 2x_2 \\ -1 + 2x_1 + 2x_2 \end{bmatrix}.$$

Thus:

$$\nabla f_1 = \nabla f(0, 0) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad S_1 = -\nabla f_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

Along this direction:

$$f(X_1 + \lambda_1 S_1) = f(-\lambda_1, +\lambda_1) = \lambda_1^2 - 2\lambda_1.$$

Derivative:

$$\frac{d}{d\lambda_1} = 2\lambda_1 - 2 = 0 \quad \Rightarrow \quad \lambda_1^* = 1.$$

Update:

$$X_2 = X_1 + S_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

Iteration 2

$$\nabla f_2 = \nabla f(-1, 1) = \begin{bmatrix} -1 \\ -1 \end{bmatrix}.$$

Norms:

$$\|\nabla f_1\|^2 = 2, \quad \|\nabla f_2\|^2 = 2.$$

Direction:

$$S_2 = -\nabla f_2 + \frac{\|\nabla f_2\|^2}{\|\nabla f_1\|^2} S_1 = \begin{bmatrix} 0 \\ 2 \end{bmatrix}.$$

Compute step:

$$f(X_2 + \lambda_2 S_2) = f(-1, 1 + 2\lambda_2) = 4\lambda_2^2 - 2\lambda_2 - 1.$$

Derivative:

$$8\lambda_2 - 2 = 0 \Rightarrow \lambda_2^* = \frac{1}{4}.$$

Update:

$$X_3 = \begin{bmatrix} -1 \\ 1.5 \end{bmatrix}.$$

Iteration 3

$$\nabla f_3 = \nabla f(-1, 1.5) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Thus:

$$\beta_3 = 0, \quad S_3 = 0.$$

No further descent direction exists, hence X_3 is the minimizer.

7 Conclusion

The Fletcher–Reeves method constructs conjugate directions using only gradient information. For quadratic functions, it converges in at most n iterations, as illustrated in the example.